1 Firm's cost minimization with CES production function

2 Define:

$X_i =$ the quantity of input i
$W_i =$ the price of input i
$Q =$ the quantity of output
$\delta_i =$ share parameter for input i
$-1 < \rho < \infty$ is parameter.

Firm's problem is to minimize cost of production subject to a given technology:

$$\min_{X_i} \sum_i W_i X_i \text{ s.t. } Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}}$$

$$\mathcal{L} = \sum_i W_i X_i + \lambda \left\{ Q - \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}} \right\}$$

First order conditions:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial X_i} &= 0 \text{ for all } i = 1, \ldots, n \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= 0 \end{aligned}$$

$$W_i - \lambda \left(-\frac{1}{\rho} \right) \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}-1} \left(-\rho \delta_i X_i^{-\rho-1} \right) = 0 \text{ for all } i = 1, \ldots, n \tag{1}$$

$$Q - \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}} = 0 \tag{2}$$

Simplify 1:

$$W_i = \lambda \delta_i X_i^{-\rho-1} \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}-1} \text{ for all } i = 1, \ldots, n$$

Ratio of two foc's (1):

$$\frac{W_i}{W_j} = \frac{\delta_i}{\delta_j} \frac{X_i^{-\rho-1}}{X_j^{-\rho-1}} \text{ for all } i = 1, \ldots, n \tag{3}$$

Solve 3 for X_i:

$$
\begin{aligned}
X_i^{-\rho-1} &= \frac{\delta_j}{\delta_i}\frac{W_i}{W_j X_j^{\rho+1}} \Leftrightarrow X_i = \left(\frac{\delta_j}{\delta_i}\frac{W_i}{W_j X_j^{\rho+1}}\right)^{\frac{1}{-\rho-1}} \qquad (4)\\
X_i^{-\rho} &= \left(\frac{\delta_j}{\delta_i}\frac{W_i}{W_j X_j^{\rho+1}}\right)^{\frac{-\rho}{-\rho-1}} \Leftrightarrow X_i^{-\rho} = \left(\frac{\delta_j}{\delta_i}\frac{W_i}{W_j X_j^{\rho+1}}\right)^{\frac{\rho}{1+\rho}}
\end{aligned}
$$

Substitute 4 into 2:

$$
\begin{aligned}
Q &= \left\{\sum_i \delta_i \left(\frac{\delta_j}{\delta_i}\frac{W_i}{W_j X_j^{\rho+1}}\right)^{\frac{\rho}{1+\rho}}\right\}^{-\frac{1}{\rho}} = \left\{\sum_i \delta_i \left(\frac{\delta_j}{\delta_i}\frac{W_i}{W_j}\right)^{\frac{\rho}{1+\rho}} X_j^{\frac{-(\rho+1)\rho}{1+\rho}}\right\}^{-\frac{1}{\rho}}\\
&= X_j \left\{\sum_i \delta_i \left(\frac{\delta_j}{\delta_i}\frac{W_i}{W_j}\right)^{\frac{\rho}{1+\rho}}\right\}^{-\frac{1}{\rho}}
\end{aligned}
$$

Solve for X_j:

$$
\begin{aligned}
X_j &= Q\left\{\sum_i \delta_i \left(\frac{\delta_j}{\delta_i}\frac{W_i}{W_j}\right)^{\frac{\rho}{1+\rho}}\right\}^{\frac{1}{\rho}}\\
&= Q\left\{\sum_i \delta_j^{\frac{\rho}{1+\rho}} \delta_i \left(\frac{1}{\delta_i}\frac{W_i}{W_j}\right)^{\frac{\rho}{1+\rho}}\right\}^{\frac{1}{\rho}} = Q\left\{\delta_j^{\frac{\rho}{1+\rho}} \sum_i \delta_i \left(\frac{1}{\delta_i}\frac{W_i}{W_j}\right)^{\frac{\rho}{1+\rho}}\right\}^{\frac{1}{\rho}}\\
&= Q\delta_j^{\frac{1}{1+\rho}}\left\{\sum_i \delta_i \left(\frac{1}{\delta_i}\frac{W_i}{W_j}\right)^{\frac{\rho}{1+\rho}}\right\}^{\frac{1}{\rho}} = Q\delta_j^{\frac{1}{1+\rho}}\left\{W_j^{-\frac{\rho}{1+\rho}} \sum \delta_i \delta_i^{-\frac{\rho}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right\}^{\frac{1}{\rho}}\\
&= Q\delta_j^{\frac{1}{1+\rho}} W_j^{-\frac{1}{1+\rho}}\left\{\sum \delta_i \delta_i^{-\frac{\rho}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right\}^{\frac{1}{\rho}} = Q\delta_j^{\frac{1}{1+\rho}} W_j^{-\frac{1}{1+\rho}}\left\{\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right\}^{\frac{1}{\rho}}\\
&= Q\delta_j^{\frac{1}{1+\rho}} W_j^{-\frac{1}{1+\rho}}\left\{\left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{-\rho}}\right\}^{\frac{1}{\rho}\frac{-\rho}{1+\rho}} = Q\delta_j^{\frac{1}{1+\rho}}\left\{\frac{W_j}{\left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}}}\right\}^{-\frac{1}{1+\rho}}
\end{aligned}
$$

$$
X_j = Q\delta_j^{\frac{1}{1+\rho}}\left(\frac{W_j}{c}\right)^{-\frac{1}{1+\rho}} \qquad (5)
$$

where c:

$$
c = \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}} \qquad (6)
$$

It can be shown that c is the unit cost of production. Define C as the total cost:

$$
\begin{aligned}
C &= \sum W_i X_i = \sum W_i Q \delta_i^{\frac{1}{1+\rho}} \left(\frac{W_i}{c}\right)^{-\frac{1}{1+\rho}} = Qc^{\frac{1}{1+\rho}} \sum \delta_i^{\frac{1}{1+\rho}} W_i^{1-\frac{1}{1+\rho}} \\
&= Qc^{\frac{1}{1+\rho}} \sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{1+\rho-1}{1+\rho}} = Qc^{\frac{1}{1+\rho}} \sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \\
&= Qc^{\frac{1}{1+\rho}} \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}\frac{\rho}{1+\rho}} = Qc^{\frac{1}{1+\rho}} c^{\frac{\rho}{1+\rho}} = Qc^{\frac{1+\rho}{1+\rho}} \\
C &= cQ \qquad (7)
\end{aligned}
$$

Define elasticity of substitution:

$$
\begin{aligned}
\sigma &= \frac{1}{1+\rho}, \text{ or} \qquad (8) \\
\sigma(1+\rho) &= 1 \Leftrightarrow \sigma + \sigma\rho = 1 \Leftrightarrow \sigma\rho = 1-\sigma \\
\rho &= \frac{1-\sigma}{\sigma}
\end{aligned}
$$

Questions:

1. Interpret the elasticity of substitution!
2. What is its relationship with *factor intensity*?

Substitute: 8 into 5:

$$
\begin{aligned}
X_j &= Q\delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c}\right)^{-\frac{1}{1+\rho}} = Q\delta_j^{\frac{1}{1+\frac{1-\sigma}{\sigma}}} \left(\frac{W_j}{c}\right)^{-\frac{1}{1+\frac{1-\sigma}{\sigma}}} = Q\delta_j^{\frac{1}{\frac{\sigma+1-\sigma}{\sigma}}} \left(\frac{W_j}{c}\right)^{-\frac{1}{\frac{\sigma+1-\sigma}{\sigma}}} \\
X_j &= Q\delta_j^{\sigma} \left(\frac{W_j}{c}\right)^{-\sigma} \qquad (9)
\end{aligned}
$$

Substitute: 8 into 6:

$$
\begin{aligned}
c &= \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}} = \left(\sum \delta_i^{\frac{1}{1+\frac{1-\sigma}{\sigma}}} W_i^{\frac{\frac{1-\sigma}{\sigma}}{1+\frac{1-\sigma}{\sigma}}}\right)^{\frac{1+\frac{1-\sigma}{\sigma}}{\frac{1-\sigma}{\sigma}}} \\
&= \left(\sum \delta_i^{\frac{1}{\frac{\sigma+1-\sigma}{\sigma}}} W_i^{\frac{\frac{1-\sigma}{\sigma}}{\frac{\sigma+1-\sigma}{\sigma}}}\right)^{\frac{\frac{\sigma+1-\sigma}{\sigma}}{\frac{1-\sigma}{\sigma}}} = \left(\sum \delta_i^{\frac{1}{\frac{1}{\sigma}}} W_i^{\frac{\frac{1-\sigma}{\sigma}}{\frac{1}{\sigma}}}\right)^{\frac{\frac{1}{\sigma}}{\frac{1-\sigma}{\sigma}}} \\
c &= \left(\sum \delta_i^{\sigma} W_i^{1-\sigma}\right)^{\frac{1}{1-\sigma}} \qquad (10)
\end{aligned}
$$

Cost shares:

$$
\begin{aligned}
S_j &= \frac{W_j X_j}{C} = \frac{W_j Q \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c}\right)^{-\frac{1}{1+\rho}}}{cQ} = \frac{W_j \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c}\right)^{-\frac{1}{1+\rho}}}{c} \\
&= \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{1+\rho-1}{1+\rho}} c^{\frac{1}{1+\rho}}}{c} = \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{cc^{-\frac{1}{1+\rho}}} = \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{c^{\frac{1+\rho-1}{1+\rho}}} = \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{c^{\frac{\rho}{1+\rho}}} \\
&= \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{\left(\left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}}\right)^{\frac{\rho}{1+\rho}}} = \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}}
\end{aligned}
$$

Cost share with σ:

$$
\begin{aligned}
S_j &= \frac{\delta_j^{\frac{1}{1+\frac{1-\sigma}{\sigma}}} W_j^{\frac{\frac{1-\sigma}{\sigma}}{1+\frac{1-\sigma}{\sigma}}}}{\sum \delta_i^{\frac{1}{1+\frac{1-\sigma}{\sigma}}} W_i^{\frac{\frac{1-\sigma}{\sigma}}{1+\frac{1-\sigma}{\sigma}}}} = \frac{\delta_j^{\frac{1}{\frac{\sigma+1-\sigma}{\sigma}}} W_j^{\frac{\frac{1-\sigma}{\sigma}}{\frac{\sigma+1-\sigma}{\sigma}}}}{\sum \delta_i^{\frac{1}{\frac{\sigma+1-\sigma}{\sigma}}} W_i^{\frac{\frac{1-\sigma}{\sigma}}{\frac{\sigma+1-\sigma}{\sigma}}}} = \frac{\delta_j^{\sigma} W_j^{1-\sigma}}{\sum \delta_i^{\sigma} W_i^{1-\sigma}} \\
S_j &= \frac{\delta_j^{\sigma} W_j^{1-\sigma}}{\sum \delta_i^{\sigma} W_i^{1-\sigma}}
\end{aligned}
$$

Illustration with 2 factors production function:

$$
Q = \left(\delta_K K^{-\rho} + \delta_L L^{-\rho}\right)^{-\frac{1}{\rho}}
$$

The unit cost:

$$
c = \left(\delta_K^{\sigma} r^{1-\sigma} + \delta_L^{\sigma} w^{1-\sigma}\right)^{\frac{1}{1-\sigma}}
$$

The factor demands:

$$
\begin{aligned}
K &= Q\delta_K^{\sigma}\left(\frac{r}{c}\right)^{-\sigma} \\
L &= Q\delta_L^{\sigma}\left(\frac{w}{c}\right)^{-\sigma}
\end{aligned}
$$

Isoquant for illustration:

$$
\begin{aligned}
Q &= \left(\delta_K K^{-\rho} + \delta_L L^{-\rho}\right)^{-\frac{1}{\rho}} \Leftrightarrow Q^{-\rho} = \delta_K K^{-\rho} + \delta_L L^{-\rho} \\
\delta_K K^{-\rho} &= Q^{-\rho} - \delta_L L^{-\rho} \Leftrightarrow K^{-\rho} = \frac{Q^{-\rho} - \delta_L L^{-\rho}}{\delta_K} \\
K^Q\left(Q, L\right) &= \left(\frac{Q^{-\rho} - \delta_L L^{-\rho}}{\delta_K}\right)^{-\frac{1}{\rho}}, \text{ or} \\
K^Q(Q, L) &= \left(\frac{Q^{-\frac{1-\sigma}{\sigma}} - \delta_L L^{-\frac{1-\sigma}{\sigma}}}{\delta_K}\right)^{-\frac{1}{\frac{1-\sigma}{\sigma}}} = \left(\frac{Q^{\frac{\sigma-1}{\sigma}} - \delta_L L^{\frac{\sigma-1}{\sigma}}}{\delta_K}\right)^{-\frac{\sigma}{1-\sigma}}
\end{aligned}
$$

Isocost:

$$
C = rK + wL \Leftrightarrow rK = C - wL \Leftrightarrow K^C(Q) = \frac{cQ - wL}{r}
$$

Factor intensity:

$$f = \frac{Q\delta_K^\sigma \left(\frac{r}{c}\right)^{-\sigma}}{Q\delta_L^\sigma \left(\frac{w}{c}\right)^{-\sigma}} = \left(\frac{\delta_K}{\delta_L}\frac{w}{r}\right)^\sigma$$

Question:

1. What is *homotheticity*?
2. What is the connection between *homotheticity* and factor demand elasticity with respect to ouput?

3 Firm's cost minimization with CES production function

$$\min_{x_i} \sum_i w_i x_i s.t. y = \prod_i x_i^{\alpha_i}$$

Define $k = \sum_i \alpha_i$ as returns to scale parameter.

$$k = \begin{cases} > 1 \Rightarrow IRS \\ = 1 \Rightarrow CRS \\ < 1 \Rightarrow DRS \end{cases}$$

Construct the Lagrange function:

$$\mathcal{L} = \sum_i w_i x_i + \lambda \left(y - \prod_i x_i^{\alpha_i} \right)$$

FOCs:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial x_i} &= w_i - \lambda \alpha_i x_i^{\alpha_i - 1} \prod_{j \neq i} x_j^{\alpha_j} = 0 \\
&= w_i - \lambda \alpha_i x_i^{-1} x_i^{\alpha_i} \prod_{j \neq i} x_j^{\alpha_j} = 0 \\
&= w_i - \lambda \frac{\alpha_i}{x_i} \prod_j x_j^{\alpha_j} = w_i + \lambda \frac{\alpha_i}{x_i} y = 0 \\
\frac{\partial \mathcal{L}}{\partial \lambda_i} &= y - \prod_i x_i^{\alpha_i} = 0
\end{aligned}$$

$$\frac{w_i}{w_j} = \frac{\lambda \frac{\alpha_i}{x_i} y}{\lambda \frac{\alpha_j}{x_j} y} = \frac{\alpha_i}{x_i}\frac{x_j}{\alpha_j}$$

To find the solution:

$$\frac{w_i}{w_j} = \frac{\alpha_i}{x_i}\frac{x_j}{\alpha_j} \Rightarrow x_i = \frac{\alpha_i x_j w_j}{\alpha_j w_i}$$

Substitute into production function:

$$
\begin{aligned}
y &= \prod_i x_i^{\alpha_i} = \prod_i \left(\frac{\alpha_i x_j w_j}{\alpha_j w_i}\right)^{\alpha_i} = \prod_i \left(\frac{x_j w_j}{\alpha_j}\right)^{\alpha_i} \prod_i \left(\frac{\alpha_i}{w_i}\right)^{\alpha_i} \\
y &= \left(\frac{x_j w_j}{\alpha_j}\right)^{\sum_i \alpha_i} \prod_i \left(\frac{\alpha_i}{w_i}\right)^{\alpha_i} = \left(\frac{x_j w_j}{\alpha_j}\right)^{k} \prod_i \left(\frac{\alpha_i}{w_i}\right)^{\alpha_i} \\
y^{\frac{1}{k}} &= \frac{x_j w_j}{\alpha_j} \left(\prod_i \left(\frac{\alpha_i}{w_i}\right)^{\alpha_i}\right)^{\frac{1}{k}} \\
x_j &= \frac{\alpha_j}{w_j} y^{\frac{1}{k}} \left(\prod_i \left(\frac{w_i}{\alpha_i}\right)^{\alpha_i}\right)^{\frac{1}{k}} \\
x_j^* &= \frac{\alpha_j c^{\frac{1}{k}} y^{\frac{1}{k}}}{w_j} \text{ where } c = \prod_i \left(\frac{w_i}{\alpha_i}\right)^{\alpha_i}
\end{aligned}
$$

Cost function:

$$
\begin{aligned}
C &= \sum_i w_i x_i^* = \sum_i w_i \frac{\alpha_i c^{\frac{1}{k}} y^{\frac{1}{k}}}{w_i} = \sum_i \alpha_i c^{\frac{1}{k}} y^{\frac{1}{k}} = c^{\frac{1}{k}} y^{\frac{1}{k}} \sum_i \alpha_i \\
C &= k c^{\frac{1}{k}} y^{\frac{1}{k}}
\end{aligned}
$$

Marginal cost:

$$
\frac{dC}{dy} = \frac{1}{k} k c^{\frac{1}{k}} y^{\frac{1}{k}-1} = c^{\frac{1}{k}} y^{\frac{1-k}{k}}
$$

Average cost:

$$
\frac{C}{y} = k c^{\frac{1}{k}} y^{\frac{1}{k}} y^{-1} = k c^{\frac{1}{k}} y^{\frac{1-k}{k}}
$$